

NELSON KANG

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Education

University of Waterloo

Bachelor's Degree, Computer Science and Finance (CFM)

- **Coursework:** Advanced Functional Programming (Racket), Data Structures & Algorithms (C), Linux/Bash/Git Tooling, Linear Algebra, Calculus II
- **Activities:** Poker Club Executive, Competitive Intramural Volleyball Captain

Experience

Backend & Data Engineering Co-op

Jun 2026 – Present

Aquanow

Toronto, ON

- Built and own a daily **credit-exposure reporting pipeline** (Python, PyIceberg, AWS Athena, S3) scanning multi-currency client balances against **Apache Iceberg** tables and converting exposure to USD via live FX rates; covered by **31 unit tests** with all AWS/Kafka dependencies mocked.
- Diagnosed and fixed **5 production bugs** in a live regulatory reporting system submitting daily trading activity to a national financial regulator, including a full-day snapshot-boundary offset and in-asset fee misattribution. Restored reconciliation from **0% to 107/107 currencies** across three reporting dates.
- Engineered an offline **reconciliation harness** (Python, DuckDB) reproducing the regulator's consistency checks against production data, and audited **10+ files** across three report jobs, surfacing silent data-loss bugs by severity.

Software Engineer Intern

Jun 2024 – Aug 2024

Ericsson Canada Inc.

Ottawa, ON

- Built **4 ETL pipelines** using **C#** and **SQL Server** to extract Jira data from JSON payloads, transform custom field mappings, and **join across multiple Jira endpoints and internal tables** to load into reporting tables; used **temporal tables** for full change-history tracking.
- Audited and catalogued all existing team pipelines, identifying and documenting faulty custom-field mappings producing **NULL data**; created a reference spreadsheet that became the team's **pipeline debugging resource**.
- Developed and maintained **Power BI dashboards** for trouble report visibility, supporting stakeholder reporting and data-driven decision-making across the BI team.

Projects

ClipFarm — Automated Volleyball Highlight Generator

Mar 2026 – Present

Python, FastAPI, Next.js, TypeScript, YOLOv8-pose, Roboflow, PostgreSQL, Docker, Cloudflare R2

- Engineered a **multi-signal highlight pipeline**: a custom-trained **Roboflow** ball detector tracks trajectory to bound rallies, **audio cheer detection** and rally-shape scoring gate highlights, and **YOLOv8-pose** classifies player actions.
- Containerized with **Docker Compose**: **FastAPI** serves auth-gated endpoints, **Celery** + **Redis** distributes jobs, and **Modal** offloads GPU inference. Auth via **Supabase JWKS** with **Row-Level Security**.
- Implemented the frontend in **Next.js** + **TypeScript** with a filterable clip grid, inline multi-label correction UI for future model training, clip trimming with background re-cut via Celery, and lazy-loaded thumbnails.

HUME — 1st Place, CORTEX BioSciences Track @ Hack the 6ix (\$1,000 + work trial)

Jul 2026

Python (standard library only), Knowledge Graphs, Adversarial Robustness, Metamorphic Testing

- Built a **deterministic belief-revision engine (no LLM)** that ingests streaming evidence into a **knowledge graph**, sizing each update in **log-odds** by evidence strength and scoping refuted claims to the condition that breaks them.
- Blocked **prompt-injection** structurally. Untrusted text cannot mutate state; writes pass a closed **delta vocabulary** with magnitude from **structured provenance** only. Independent **~90-attack red-team: zero** text-driven mutations.
- Locked extraction behind one audited seam with **19 probe suites: metamorphic tests** (paraphrase, entity renaming, negation) plus an **AST build guard**, catching two silent false-negative regressions.

Multi-Factor Robo-Advisor — 1st Place, Market Beat Competition

Nov 2025

Python, Pandas, NumPy, yfinance

- Optimized a **\$1M CAD portfolio** across U.S. and Canadian equities; achieved **5.7% return** vs. 3.7% market average benchmark, placing **1st out of 18 teams**.
- Implemented a **multi-factor portfolio construction pipeline** applying momentum, volatility, correlation, and trend signals inspired by the **Ranked Asset Allocation Model (RAAM)**.
- Automated end-to-end: data ingestion, liquidity screening, factor ranking, **sector-cap enforcement**, and **Sharpe-ratio optimization** under real-world constraints (transaction fees, FX conversion, position limits).